Exploring the Effects of the 2005 Arkansas Online Ban of Tobacco Sales

Colin Donahue

Advanced Econometrics

1 May 2022

1 Introduction

The National Center for Disease Control (CDC) lists tobacco and nicotine use as “the leading cause of preventable disease, disability, and death in the United States” (CDC 2022). Since the introduction of anti-smoking campaigns like the TRUTH Initiative, the largest nonprofit committed to combating tobacco and nicotine addiction, it has been well-known that tobacco use is linked to negative health outcomes. While smoking rates in both adults and children have declined since 1998, the mass introduction of flavored e-cigarettes, a nicotine delivery system considered a “healthier alternative” to traditional tobacco products, has created new concerns about youth nicotine addiction (ALA 2018). In 2015, the e-cigarette company JUUL rose to prominence by marketing flavored e-cigarettes to consumers (Business Insider 2022). The market for flavored e-cigarettes has become popular amongst younger generations and contributed to high teenage nicotine addiction rates. In fact, “flavors play a big role in the youth e-cigarette epidemic — 97% of youth who vape use flavored products” (TRUTH 2022).

While the casual health consequences of e-cigarettes remain unknown, there is substantial evidence to suggest that these products are harmful to lungs and brain development. In 2020, the CDC identified over 2,800 incidents of vaping-associated lung injuries (EVALI) and reported over 68 deaths (CDC 2020). In addition to negative health consequences, youth e-cigarette consumers are also more likely to switch to tobacco products in the future: “Young people who had ever used e-cigarettes had seven times higher odds of becoming smokers one year later compared with those who had never vaped” (TRUTH 2020). The U.S. has a strong incentive to stop the e-cigarette epidemic. Currently, the U.S. government spends more than $225 billion annually on medical care expenses to treat smoking-related addiction and illnesses (CDC 2022). In response to the emerging youth nicotine epidemic, the Federal Department of Agriculture implemented a federal ban on the sale of flavored e-cigarette products to discourage underage use. The government further passed the T21 ACT in 2019, a premiere policy that federally changed the legal age to purchase nicotine to 21. While these two policies were intended to reduce youth nicotine use, they fail to fully consider other means of youth access to these products. A recent public health report published by the Mitchell Hamline School of Law in Minnesota explains that underage consumers may still be able to order nicotine products online due to a lack of age verification systems: “the public health community has a new reason to address illegal online sales of tobacco to youth and young adults” (Hamline Law 2022).

While there are currently 33 U.S. states that regulate online tobacco and nicotine sales, many of these policies fail to verify the age of consumers online. It would be of interest to policymakers to determine if a complete online sale and delivery ban of nicotine products is needed to reduce youth nicotine use. There have only been four states—New York, Maryland, Connecticut, and Arkansas—in the United States that completely banned online tobacco product sales altogether. Interestingly, there is little to no research on the effects of these four online business-to-consumer sale bans of tobacco products. These policies would be useful to study to help understand if the 2021 Preventing Online Sales of E-Cigarettes to Children Act (POSECCA) should be amended to completely ban the online sales and delivery of nicotine products, like e-cigarettes. Using data from the Youth Risk Behavior Surveillance System (YRBS) between 2001 and 2007, I use difference-in-difference (DD) models to determine the effects of the policy on youth tobacco consumption. Consistent with current research on the impact of tobacco policies at the state-level, multiple control groups are used to estimate DD models. In additional specifications, DD models are estimated using data on adult consumers from The Tobacco Use Supplement to the Current Population Survey (TUS-CPS) to determine if the policy impacted legal-age consumers. In theory, this group should not be largely impacted by the policy given that they can legally buy these products in stores.

2 Background

To approach this topic, the paper is organized in the following manner. There is a review of current research on state and federal initiatives aimed to reduce youth access to tobacco and nicotine products. This review helps contextualize this paper’s contribution to existing literature. Next, I list the data sources used for the analysis in addition to an empirical methodology. Limitations to the data and methodology are discussed, and result interpretations are presented.

2.1 Youth tobacco policies in the United States

There have been many policies and initiatives aimed to reduce youth access to tobacco products. Between 1995 and 2007, 33 states attempted to regulate the online sale and delivery of tobacco products. Most of these policies, however, did not overcome challenges to online age verification systems (Hamline Law 2022). All but 4 states—Arkansas, New York, Connecticut, and Maryland—completely banned the online sale and delivery of tobacco products by implementing strict face-to-face purchase requirements. In 2009, the U.S. passed the federally mandated Prevent All Cigarette Trafficking Act of 2009 (PACT Act), which attempted to restrict online purchases of tobacco products. While many states implemented similar regulations to Arkansas, New York, Connecticut, and Mayland after the 2009 policy, the majority of states continue to allow online tobacco retail sales. It is important for policymakers to understand if an online ban is needed to reduce youth nicotine use.

Today, age verification systems have become weaker than ever. In a recent controlled experiment completed by the University of North Carolina, participants aged 11 to 15 years old were instructed to purchase tobacco products using the internet under the supervision of researchers. Of 83 attempts to purchase tobacco products in 12 states, 92% were successful and delivered without requiring a signature (Williams et al. 2017). This is particularly concerning given that the internet has become increasingly accessible to younger generations. In addition, new delivery app services, such as GoPuff in Philadelphia which connects self-employed delivery drivers to consumers, allow nicotine products to be delivered without systematic age verification systems (Valada 2021). Online tobacco sales continue to be a viable method for underage users to illegally purchase tobacco and nicotine products. These loopholes threaten the validity and intentions of other other policies.

At present, there have been no studies that have evaluated the effects of banning online sales of tobacco products in Arkansas, New York, Connecticut, or Maryland. It is difficult to know if these policies would actually decrease youth nicotine consumption or if youth consumers would find new means of purchasing or obtaining these products. In addition, since e-cigarettes are theoretically a healthier alternative for people trying to quit vaping, understanding how adult consumers are affected by this policy would be relevant. A federal online ban might do harm to individuals trying to quit a nicotine addiction in locations that do not have access to stores that sell nicotine. To maximize welfare, policy makers would hope that an online ban would reduce youth nicotine consumption without affecting adults trying to quit their addiction.

2.2 Youth tobacco consumption research in United States

Online tobacco and nicotine sales, including e-cigarette products, make up a non-trivial amount of sales. According to a 2021 market research report, about 22% of current e-cigarette sales come from online retailers and/or delivery platforms (Grand View Research 2022). Despite improvements in tobacco policies in recent years, it is unclear which policies affect youth tobacco consumption because multiple laws have been passed at the same time. Most tobacco policy research primarily focuses on the years prior to 2017 due to the large number of simultaneous effective dates of state and federal policies in addition to a lack of contemporary available data. At present, there are a limited number of empirical studies looking at policies that affect youth tobacco and nicotine consumption.

In the past, researchers have attempted to determine how credit card company commitments to block online tobacco purchases affected youth tobacco consumption. In 2005, many credit companies like PayPal made voluntary, non-legal agreements to block their credit services from being used for online tobacco sales. Ribisl et al. found that the top 50 online tobacco retailers across the US received significantly less consumer traffic after these commitments when controlling for increasing US internet traffic over time (2011). However, this study only looks at aggregate decreases in *visits* to online tobacco retailers and fails to consider if these commitments actually decreased youth tobacco purchases or consumption. In addition, since these commitments occurred at the national level, this would be a shock to all states rather than a causal analysis looking at the difference between treated and non-treated states. The aggregated decrease in visits to online tobacco retailers cannot be separated from the already declining tobacco use rates between 1998-2007. Moreover, there is little information regarding whether these companies actually followed their commitments.

Researchers have also looked at the effects of increasing the minimum purchase age of tobacco to 21. Hawaii and California, two states that changed the legal tobacco purchase age in 2016, showed statistically significant decreases of 12.1% in youth cigarette use in Hawaii and 7.1% in California (Jenco 2020). Other researchers have looked at unique cities that raised the tobacco purchase age to 21. Farley and Jasek found that the increase in the tobacco purchase age to 21 in New York City in 2011 and Chicago in 2016 were associated with a 29% and a 56% decrease in tobacco use in high school students respectively (2017).

In addition to tobacco 21 laws, some researchers have evaluated the effects of unique flavored nicotine ban in unique states. Friedman et al. used a difference-in-difference approach to study San Francisco's 2016 nicotine flavor ban on tobacco consumption in young adults (2021). Using data from YRBS, Friedman et al. found that the flavor ban was associated with no statistically significant decrease in smoking among minor high school students when compared to non-treated districts (2021). These results were particularly concerning because they may indicate that the 2020 Federal Flavor Ban may push underage users to substitute away from flavored vapes towards traditional tobacco products, which is theoretically a more unhealthy alternative. Due to the lack of studies that address youth access, this study aims to evaluate Arkansas’s 2005 ban on direct-to-consumer sales of tobacco products. In particular, I follow the differences-in-differences used by Friedman et al. to estimate the effect of the passage of the online ban on tobacco consumption with individuals under the age of 18. Since the internet has become more prevalent since 2005, results that suggest a reduction in tobacco consumption may be a lower bound estimate for a contemporary policy today.

3 Data

3.1 Data descriptions

Data for this analysis were taken from the Youth Risk Behavior Surveillance System (YRBS), which is a biennial survey conducted by the Centers for Disease Control and Prevention. YRBS is an anonymous, self-administered survey completed in the spring by school districts and collects behavioral health information. YRBS is specifically designed to obtain a nationally representative sample of high school students between the ages of 12 and18. This data set is cross-sectional and contains details about cigarette and other tobacco product consumption in addition to demographic characteristics, such as age, sex, grade, and race. Data is collected using a two-stage, cluster sample design. Schools are first randomly selected based on a probability proportional to enrollment size, and then classes within each school are selected randomly. The CDC provides weights that correct for non-responses and any unequal distribution of demographic characteristics. This study only utilizes data from the period 2001-2007, with surveys from the years 2001, 2003, 2005, and 2007. Survey data from the 2009 wave could not be included due to the 2009 Family Smoking Prevention and Tobacco Control Act (TCA), which inspired many state-level unique tobacco regulation policies, including in Arkansas.

While there were four treated states—Arkansas, Maryland, New York, and Connecticut—with the same online tobacco ban policy, which would naturally lend itself to a stacked difference-in-difference approach, New York, Maryland, and Connecticut did not participate in YRBS or have complete data before 2005. To ensure there is data for both pre-policy and post-policy periods for treated states, this analysis is restricted to Arkansas. These other treated states are therefore dropped from the data set. Arkansas being the only treatment state is, therefore, a significant limitation to this study. There are no other known events during this time period that would significantly affect tobacco consumption in Arkansas. However, additional robustness checks, which are described later, are needed to ensure that there were no within-state changes in Arkansas that may influence tobacco and nicotine consumption.

For the main analysis of this policy, I follow Friedman et al. and restrict the sample to students younger than eighteen years old who had non-missing YRBS data. In additional specifications, tobacco consumption is explored in legally aged consumers over the age of eighteen. It is reasonable to expect that legal consumers would not be affected as much as underage consumers since they can legally purchase these products at a store. Data legal aged consumers were taken from the Tobacco Use Supplement in the Current Population Survey (TUS-CPS), which is a similar survey to YRBS and sponsored by the National Cancer Institute. TUS-CPS contains cross-sectional data on the same tobacco use outcome variables and demographics at the state level. Unfortunately, YRBS and TUS-CPS are incompatible and the two data sets could not be harmonized due to weighting and sample collection differences. These two analyses are therefore run separately. This analysis uses two main measures of tobacco consumption to determine the potential changes in tobacco use after the 2005 policy. The first outcome variable of interest is “any cigarette use,” a binary variable for if the student used any cigarettes in the last thirty days. The second main outcome of interest is “any smoker”, which is a binary variable for if the student smokes more than 1 cigarette per day over the last thirty days. There is an additional variable in the data set for “heavy smoker,” which is a binary variable for if the high school student smokes more than 10 cigarettes per day.

3.2 Data limitations

There are many limitations to this data set that must be acknowledged. First, data from the YRBS survey is self-reported, and behaviors may be under or over-reported. Second, YRBS was only administered in high schools and results are therefore not generalizable to the entire youth population. Third, only 44 of all 50 states are represented in the data set. There are three states—Minnesota, Oregon, and Washington—that do not participate in the YRBS survey and an additional three states—Massachusetts, Ohio, and Indiana—that do not make their data publicly available. Despite these shortcomings, YRBS is still one of the most common data sets for studying trends in youth tobacco consumption. Finally, YRBS only contains age, race, sex, and grade demographic characteristics. YRBS does not contain information on other important factors like income or proximity to a city; however, income and proximity to city are included in the TUS-CPS data set.

4 Methods

4.1 Difference-in-Difference Model

For this analysis, I look at changes in youth tobacco consumption among individuals under the legal age of eighteen in Arkansas (the treatment group) with individuals under the age of eighteen in states which are unlikely to be affected by Arkansas online tobacco sale ban (the control groups). The following equations presents the DD model for this study:

$$Y_{ist} = \beta_0 + \beta_1(ARK_{ist}) + \beta_2(POST_t) + \delta_{DD}(POST_t)*(ARK_{is}) + \beta_3 X_{it} + \mathcal{E}_{ist}\ , \tag{1}$$

where Y_{ist} is an indicator that equals one if the individual smoked cigarettes in the last thirty days in year t and state s. In additional specifications, other outcome variables for Y_{ist} are used in the same model such as indicator variables for if the individual is a heavy smoker[1] or a light smoker[2]. ARK_{ist} is an indicator for whether the individual lives in Arkansas, and $POST_t$ equals one if the survey data is collected after the policy change (2005-2007). The estimate that indicates the average treatment effect of the Arkansas online tobacco ban is δ_{DD}. The δ_{DD} estimate is the intent-to-treat (ITT) effect. Due to

[1]Heavy smokers are defined as people that smoke 10+ cigarettes per day
[2]Light smokers are defined as people that smoke less than 10 cigarettes per day

imperfect compliance, many individuals may find alternative means of obtaining tobacco or tobacco retailers may not comply with the law. This analysis only captures the causal effect of being assigned to treatment. X_{it} represents individual controls including age, sex, race, and grade (9th-12th). The model accounts for state and year fixed effects, which can address unobserved differences in tobacco consumption across time and states. All YRBS regressions are clustered by state and use YRBS weights, and all TUS-CPS regressions are clustered by county and include TUS-CPS provided weights.

Considering previous research evaluating policies with only one treatment state (Lenhart 2021), I estimate the DD model with three different control groups. Similar results in sign, magnitude, and significance across the control groups may serve as a robustness check. The first control group utilizes all states that participated in YRBS. The second control group includes Arkansas's neighboring states (Texas, Oklahoma, Missouri, Tennessee, Mississippi, and Louisiana), and this control group may be especially helpful if there are environmental factors or cultural norms that contribute to tobacco consumption. There appears to be some evidence of neighboring states being a good control in Figure 3 in the Appendix, where states bordering Arkansas have similarly high tobacco use rates and could serve as an effective counterfactual. In addition, the third control group consists of states that were determined to be most similar to Arkansas (Hawaii, New Jersey, Nevada, Rhode Island, New Mexico, New Hampshire, and Texas) as determined by the Daily Kos Elections State Similarity Index, a credible resource that uses geographic size, population, races, poverty rates, and other demographics to identify states that are theoretically the most similar (Jarman 2020). Given the relatively small number of clusters, I employ a cluster bootstrap resampling method due to a concern brought upon by Bertrand et al., who showed that the rejection rates can significantly increase as the number of clusters decreases (2003). Clustered bootstraps may help combat over rejection rates and provide more conservative standard errors (Horowitz 2019). In addition, sensitivity to sampling weights are explored due to the relatively limited number of clusters.

4.2 Parallel Trends Assumption

The most important identification assumption for the DD model is parallel trends, where the difference between the control group and the treatment group remains constant during the pre-period. I graph the “pre” and “post” trends of youth tobacco consumption in Arkansas and further use an event study analysis to look at the pre-trend. The event analysis uses years 1999 - 2009 with a reference year of 2001. Theoretically, there is reason to believe that the parallel trends assumption should hold. As shown in Table 1, 2001-2007 is a unique time frame where there were no other known state or federal policies or events that would affect tobacco consumption in Arkansas.

It is important to acknowledge that three of the largest mail delivery services in the US—Federal Express (FedEx), United Parcel Service (UPS), and Dalsey Hillblom Lynn (DHL)—committed to prohibiting shipments of tobacco products directly to consumers in 2005 (CA Group 2018). These commitments to stop shipping tobacco products would be a shock to all control states and therefore not violate the parallel trends assumption. The United States Postal Service (USPS), which is the largest delivery service in the United States run by the government, also did not ban tobacco shipments. It is therefore reasonable to not expect a decrease in tobacco consumption in control states from these commitments. There are two potential concerns, however, to the parallel trends assumption. In 1998, the TRUTH Initiative was a heavily funded anti-smoking campaign designed to reduce the number of adult and youth smokers in the United States. One limitation of this study is that this campaign may have been more efficacious in states with higher baseline smoking rates like Arkansas, which has one of the highest smoking rates in the country during this period.

4.3 Robustness Checks

The first robustness check for this study is to estimate the effect of the policy on legal consumers using equation 1 using adult TUS-CPS data. One would expect that these DD estimates for individuals over 18 should have a smaller, statistically insignificant magnitude than the analysis of youth tobacco consumption since these consumers can purchase tobacco in stores. Family income, marital status, and proximity to city variables that are only available in TUS-CPS are added to X_{it} (controls) in the regression. Family income may be an important control because individuals with higher income may have access to more resources like drug and alcohol education. Second, the small number of clusters may still result in over rejection regions. To address this concern in addition to using the clustered bootstrap method for standard errors, I collapse the state-level data into averaged pre and post-periods and re-estimate the equation (1). This solution was proposed by Bertrand et al. and may help prevent over rejection rates (2003). This solution has low power and only works for laws that were passed at the same time for all treated states, which in this study is true.

4.3 Descriptive statistics

Descriptive statistics from youth participants in YRBS for the treatment group and the three control groups are presented in Table 2. The demographics for age, sex, and grade are relatively balanced across all control groups and the treatment group. It is interesting to note that Arkansas and the first two control groups have a higher percentage of white respondents.

In addition, Table 3 shows that youth tobacco consumption outcome variables in both the treatment and control groups decreased in the post-period after the policy was effective. The magnitude of this appears to be larger for Arkansas, providing some inclination of a treatment effect. Table 3 further shows that Arkansas has the highest youth tobacco use compared to the control groups.

Descriptive statistics from adult participants for the treatment group and the three control groups are presented in Table 4. The demographics for age, sex, and grade are balanced across all groups. One concern from this table is that 33.8% of the Arkansas population lives in urban areas than the populations in control groups 1, 2, and 3. This may indicate that this policy could negatively affect access to legal-aged consumers that live in remote locations that do not have physical access to tobacco stores.

Table 5 shows adult tobacco consumption outcome variables in the TUS-CPS data. Daily cigarette use unexpectedly increased in Arkansas post-policy but decreased for control groups 1, 2, and 3. All of the other outcome measures are relatively similar pre and post-period, signifying that there may not be a significant change in consumption for adults after the policy.

5 Results

5.1 Main results

Table 6 and 7 highlight the main results of this study. Table 6 looks at the percentage of youth students that smoke cigarettes, which were the most popular tobacco products at the time. All DD estimates in Table 6 Panel A show a reduction in the percentage of daily youth cigarette smokers after the policy implementation. These results are not sensitive to weighting or controls and are fairly consistent across control groups in both sign, magnitude, and significance. Column (2) implies that there was a 3.41

percentage point decrease in the frequency of daily youth cigarette smokers in Arkansas compared to control group 1. Panel B shows a similar reduction in the frequency of cigarette consumption after the policy. The estimates are larger in magnitude for “any” cigarette use than “daily” cigarette use. Intuitively, this makes sense since daily cigarette smokers are more likely to be addicted and therefore would resort to other means of obtaining cigarettes out of necessity. It is important to be cautious when interpreting the statistical significance levels here. Since there are a limited number of clusters in this sample, the results are likely to be overestimating the significance of the policy change.

Table 7 highlights alternative DD estimates that focus at the type of smoker—heavy smokers and any smokers. The results in Panel A and B are consistent with Table 6 and show percentage point reductions in youth heavy smokers and any smokers. There was a smaller percentage point decrease in “heavy” smokers than “any” smokers. This is consistent with the aforementioned idea that individuals that are addicted may be less affected by the 2005 online tobacco ban than individuals that are more casual tobacco users. Estimates across treatment and control groups in table 7 are also similar and not particularly sensitive to the weighting or demographic controls.

To better understand the effects of this policy, it is important to explore adult tobacco consumption. Table 8 provides evidence that adults were not substantially affected by the policy as much as youth. When clustered at the county level, the reduction in the percentage of daily adult smokers was not statistically significant for all three control groups. These results suggest that adult consumers that are addicted do not heavily rely on online distributors to buy tobacco products, meaning this policy would not cause harm to that population. Interestingly, there was a statistically significant 2.01 percentage point decrease in the percentage of adults who smoke anything, when controlling for demographics. While these results are significant, it is interesting to note that the magnitudes of the DD estimates are smaller than the magnitudes of DD estimates for youth.

5.2 Robustness Checks

Figure 2 and 3 in the appendix provide some evidence that the parallel trends assumptions hold. Control group 1 has a relatively similar pre-period trend to Arkansas before the implementation of the policy (policy effective date in blue). There is a large decrease in the percentage of youth smokers in Arkansas after the policy relative to control group 1. Control group 2 has a somewhat similar pre-trend period, but Arkansas has a clear steeper drop in the percentage of youth smokers relative to its neighboring states. Control group 3 satisfies the parallel trends assumption only in the pre-period 2003 to 2005. One reason for this violation may be due to one state, Louisiana, that had a large increase in tobacco consumption. The event study analysis further shows similar pre-period trends for the treatment and three control groups.

Due to the limited number of clusters in the sample, Table 9 shows DD estimates with averaged state-level data into pre and post-periods. The DD estimates for “daily” smoking and “any” smoking are less significant than the main analysis results in Table 6. For instance, control groups 2 and 3 no longer have statistically significant reductions in cigarette usage when controlling for demographics, year fixed effects, and state fixed effects. Column (2) provides some evidence that this policy was associated with a 3.08 percentage point decrease in the percentage of high school youth that smokes daily when controlling for demographics. However, it is difficult to conclude since this method has low power.

Table 10 and 11 shows DD estimates that look at the effect of the policy on the time-varying control variables used for the youth and adults analysis. The significance of the treatment effects on all of the control variables in Table 10 and most of the control variables in Table 11 suggests that this policy was likely associated with other time-varying elements that affect tobacco consumption. These tables make the results of this study less robust because it is still difficult to ensure that the effect was not due to other determinants of tobacco consumption.

Table 12 highlights another measure for youth tobacco consumption and looks at all tobacco product consumption aggregated together. The results in Panel A and B are consistent with Table 6 and show similar magnitude reductions in youth tobacco product use. This table provides evidence that the decrease in cigarette use was not accompanied by an increase in other tobacco products.

6 Conclusion

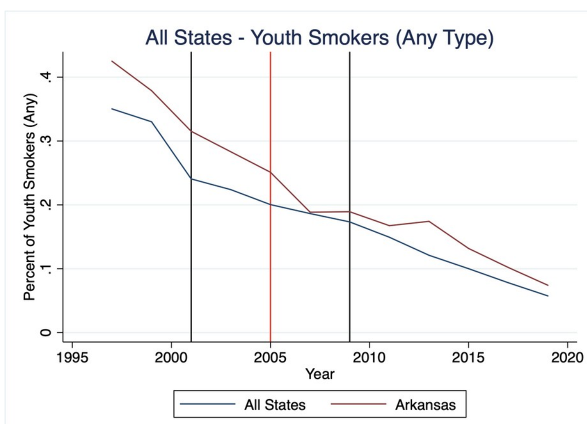
The purpose of this study was to determine the effects of the 2005 Arkansas ban of online tobacco product sales on youth nicotine consumption. This was one of the first policies aimed to address youth tobacco access and has not readily been studied. This research is especially relevant today because it may help identify whether or not the United States should ban online sale of e-cigarettes. The results of this analysis suggest that this policy was associated with a 6.67 percentage point reduction in the percentage of high school youth that smoke any amount of cigarettes and a 3.41 percentage point decrease in the percentage of high school youth that smoke daily in Arkansas compared to all states (control group 1) when controlling for demographic characteristics. If results are robust, this implies that casual smokers are affected more by this policy than addicted smokers. In addition, the magnitude of the reduction in cigarette use is larger for youth consumers than adult consumers, which provides a good argument for implementing an online ban since legal-aged consumers are not as affected by the policy.

There are many limitations to this study. The YRBS data set neither contains rich information about demographics nor complete data for all states and periods. The lack of controls and small number of clusters most likely have increased the rejection regions for the DD estimates and biased the estimates. While YRBS is one of the most frequently used data sets for tobacco policy research, this data set makes it difficult to prove causation, and several robustness checks are needed. As illustrated in the robustness check section, these results are not particularly robust but show some inclination that there was a decrease in youth tobacco consumption after the year 2005. Whether or not this was due to the online ban remains probable, but there is no complete evidence to prove causation. Future work may be able to be completed using county-level data, if it exists, or explore outcome variables related to the topic such as if an individual attempted to quit smoking.

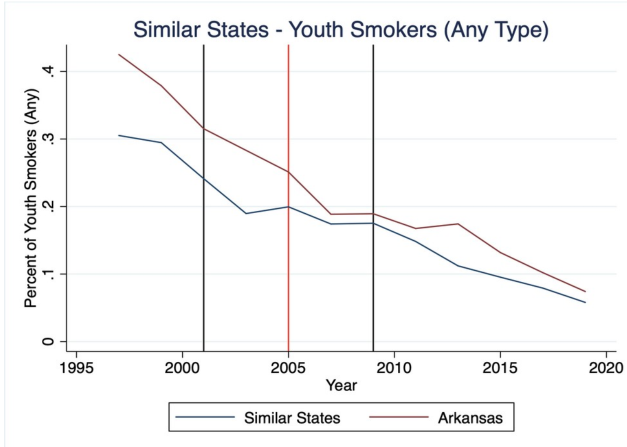
Appendix

Figure 2 Percentage of youth that smoke anything (2001-2009)

A. Control Group 1



A. Control Group 2



A. Control Group 3

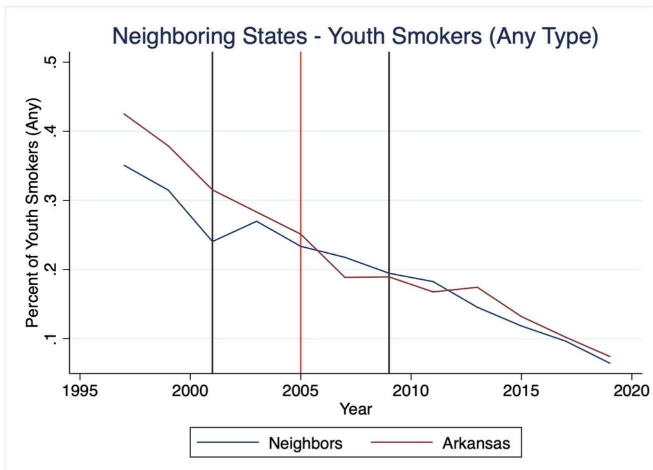
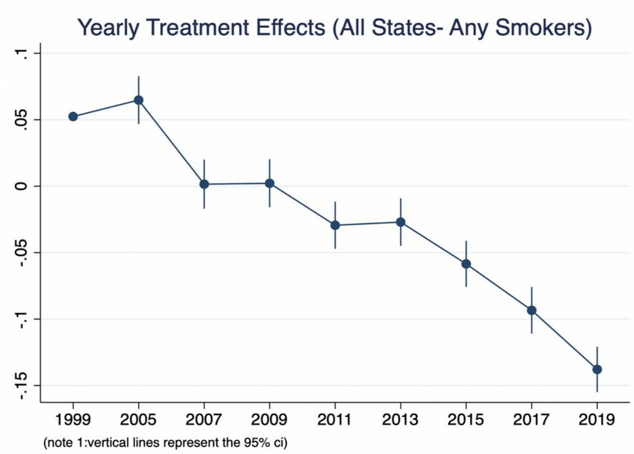
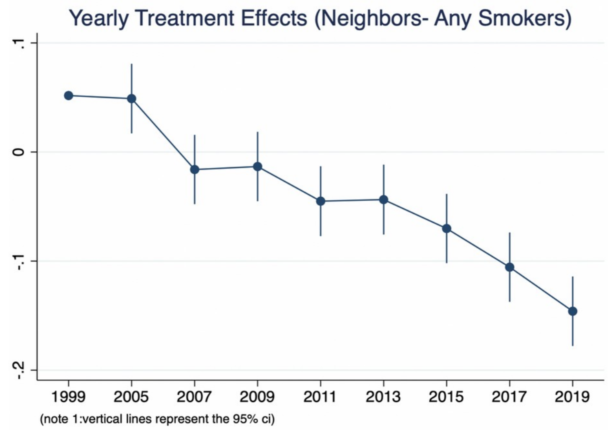


Figure 2 Event Study

A. Control Group 1



A. Control Group 2



A. Control Group 3

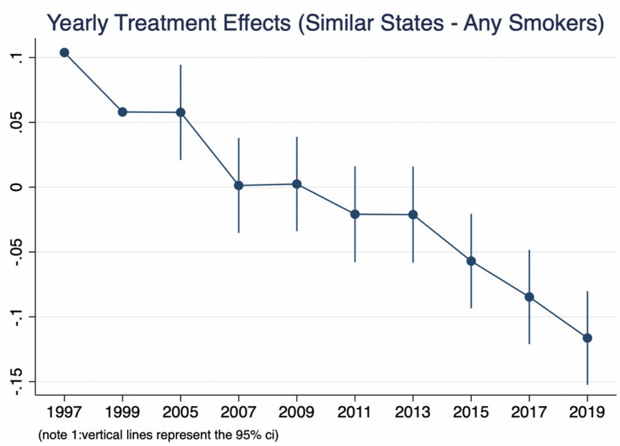
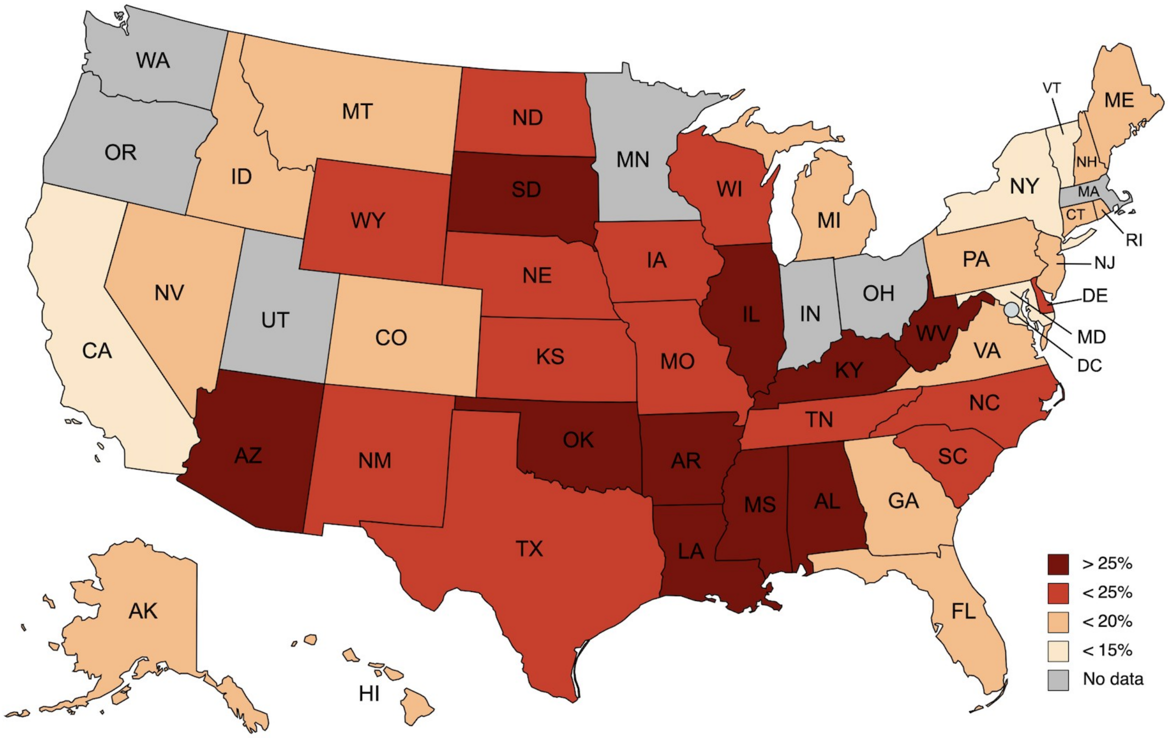


Figure 3 Youth Smoking Rates 2005 by State (YRBS Data)



<u>Citations</u>

ALA. "Trends in Cigarette Smoking Rates." American Lung Association, 2018,

www.lung.org/research/trends-in-lung-disease/tobacco-trends-brief/

overall-tobac

co-trends. Accessed 13 Apr. 2022.

Bertrand et al. "How Much Should We Trust Differences-In-Differences Estimates?" *National*

Bureau of Economic Research, June 2003.

Business Insider. "The Rise And Fall Of Juul | Rise And Fall." Youtube, 2 Feb. 2022,

www.youtube.com/watch?v=wkQ4WCAG38I&t=1s. Accessed 14 Apr. 2022.

CA Group. "FedEx Clamps Down on Cigarette Delivery." *Consumer Affairs*, 2018,
www.consumeraffairs.com/news04/2006/02/tobacco_fedex.html. Accessed 11 Apr. 2022.

CDC. "Smoking & Tobacco Use." Center for Disease Control, edited by CDC, 2022,
www.cdc.gov/tobacco/data_statistics/index.htm. Accessed 5 Apr. 2022.

CDC. “Outbreak of Lung Injury Associated with the Use of E-Cigarette, or Vaping, Products.”
25 Feb. 2020
,www.cdc.gov/tobacco/basic_information/e-cigarettes/severe-lung-
disease.html. Accessed 27 Apr. 2022.

Farley, Shannon, and John Jasek. "Evaluating Sensible Tobacco Enforcement and Tobacco 21
Laws in New York City and Chicago." *Bureau of Chronic Disease Prevention and*
Tobacco Control, 2017.

Friedman, Abigail. "A Difference-in-Differences Analysis of Youth Smoking and a Ban on Sales
of Flavored Tobacco Products in San Francisco, California." *Jama Network of Pediatrics*,
24 may 2021.

Grand View Research. "U.S. E-cigarette And Vape Market Size, Share & Trends Analysis."

Market Analysis Report, Mar. 2022.

Hamline Law. "Online sales of e-cigarettes & other tobacco products." Public Health Law Center. 1 Dec. 2019, www.publichealthlawcenter.org/sites/default/files/resources/Online-Sales-E-Cigarettes-Other-Tobacco-Products.pdf. Accessed 15 Apr. 2022.

Jarman, David. "How similar is each state to every other? Daily Kos Elections' State Similarity Index will tell you." *Daily KOS*, Feb. 2020, www.dailykos.com/stories/2020/2/19/1917029/-How-similar-is-each-state-to-every-other-Daily-Kos-Elections-State-Similarity-Index-will-tell-you. Accessed 5 Apr. 2022.

Horowitz, Joeel. “Bootstrap Methods in Econometrics.” *Annual Review of Economics,* 2019 11:1, 193-224

Jenco, Melissa. "CDC reports confirm benefits of raising tobacco age to 21." *American Academy of Pediatrics*, 16 Jan. 2020.

Lenhart, Otto. "The effects of paid family leave on food insecurity—evidence from California." *Review of Economics of the Household volume*, 2021.

Ribisl, Kurt. "Effectiveness of State and Federal Government Agreements with Major Credit Card and Shipping Companies to Block Illegal Internet Cigarette Sales." *PLOS One*, 14 Feb. 2011.

TRUTH Initiative. "Local restrictions on flavored tobacco and e-cigarette products." TRUTH Initiative, 14 Apr. 2022, truthinitiative.org/research-resources/emerging-tobacco-products/local-restrictions-flavored-tobacco-and-e-cigarette. Accessed 22 Apr. 2022.

TRUTH Initiative. "Young people who vape are much more likely to become smokers, new research confirms." TRUTH Initiative, 16 Sept. 2020,truthinitiative.org/research-resources/emerging-tobacco-products/young-people-who-vape-are-much-more-likely-become. Accessed 12 Apr. 2022.

Valada, Nick. "Drexel students' goPuff app brings smoking, snacking convenience to your door." The Inquirer, 20 Apr. 2021, www.inquirer.com/philly/blogs/lifestyle/Drexel-students-goPuff-app-brings-smoking-snacking-convenience-to-your-door.html. Accessed 19 Apr. 2022.

Williams, R.S., Derrick, J., Richardson, A., Liebman, A. K., Ribisl, K. M. (2017). Content analysis of age verification, purchase, and delivery methods of 283 internet e-cigarette vendors, 2013 and 2014. Tobacco Control, published online first: 08 May 2017. doi: 10.1136/tobaccocontrol-2016-053616.